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<https://github.com/252310057-boop/algoritma-semster2.git>

```
Jarak terpendek dari 1 ke 5 adalah: 60
Rute: 1 -> 4 -> 3 -> 5

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Process exited after 0.2485 seconds with return value 0
Press any key to continue . . .
```

```
1  #include <iostream>
2  #include <climits>
3
4  using namespace std;
5
6  const int V = 5;
7
8
9  int minDistance(int dist[], bool visited[]) {
10     int min = INT_MAX;
11     int min_index = -1;
12
13     for (int v = 0; v < V; v++) {
14         if (!visited[v] && dist[v] <= min) {
15             min = dist[v];
16             min_index = v;
17         }
18     }
19
20     return min_index;
21 }
22
23
24 void printPath(int parent[], int j) {
25     if (parent[j] == -1) {
26         cout << j + 1;
27         return;
28     }
29
30     printPath(parent, parent[j]);
31     cout << " -> " << j + 1;
32 }
33
34
35 void dijkstra(int graph[V][V], int src, int dest) {
36     int dist[V];
37     bool visited[V];
38     int parent[V];
39 }
```

```

40 for (int i = 0; i < V; i++) {
41     dist[i] = INT_MAX;
42     visited[i] = false;
43     parent[i] = -1;
44 }
45
46 dist[src] = 0;
47
48 for (int count = 0; count < V - 1; count++) {
49     int u = minDistance(dist, visited);
50     visited[u] = true;
51
52     for (int v = 0; v < V; v++) {
53         if (!visited[v] && graph[u][v] && dist[u] != INT_MAX && dist[u] + graph[u][v] < dist[v]) {
54             parent[v] = u;
55             dist[v] = dist[u] + graph[u][v];
56         }
57     }
58 }
59
60 cout << "Jarak terpendek dari " << src + 1 << " ke " << dest + 1 << " adalah: " << dist[dest] << endl;
61 cout << "Rute: ";
62 printPath(parent, dest);
63 cout << endl;
64 }
65
66 int main() {
67
68     int graph[V][V] = {
69         {0, 10, 0, 30, 100},
70         {10, 0, 50, 0, 0},
71         {0, 50, 0, 20, 10},
72         {30, 0, 20, 0, 60},
73         {100, 0, 10, 60, 0}
74     };
75
76     dijkstra(graph, 0, 4);
77
78     dijkstra(graph, 0, 4);
79     return 0;
80 }
81

```